

Table 1: BABILong ablation results for `tiiuae/falcon-mamba-7b-instruct` under the original no-few-shot StateCache scaffold. Accuracy is exact-match accuracy (%) averaged over QA1–QA3 with 50 examples per task and context length. The scaffold-only row uses the same inserted `[CACHE]` tokens but does not apply the StateCache module.

Method	0k	1k	2k	4k	8k	16k	Avg.
Scaffold only, no StateCache	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Loss-selected, $k = 1$, L62	22.7	11.3	20.0	16.0	20.0	14.7	17.4
Loss-selected, $k = 4$, L62	16.7	18.0	15.3	12.7	14.7	14.7	15.3
Loss-selected, $k = 8$, L62	20.0	18.7	15.3	21.3	16.7	22.0	19.0
Loss-selected, $k = 16$, L62	18.0	12.7	22.7	11.3	20.0	13.3	16.3
Loss-selected, $k = 32$, L62	23.3	10.0	19.3	12.7	18.7	19.3	17.2
Loss-selected, $k = 64$, L62	15.3	18.7	25.3	18.0	14.0	21.3	18.8
Loss-selected, $k = 4$, L62, train up to 8k	20.0	14.7	16.7	15.3	13.3	15.3	15.9
Loss-selected, $k = 4$, L62, no gate	18.0	15.3	14.0	14.7	10.7	16.7	14.9
Loss-selected, $k = 4$, L62, full loss	21.3	18.7	20.7	9.3	16.7	19.3	17.7
Loss-selected, $k = 4$, L32	48.7	44.7	33.3	31.3	25.3	19.3	33.8
Loss-selected, $k = 4$, L2	62.0	56.7	48.0	35.3	38.0	24.7	44.1
Uniform interval, $k = 4$, L62	21.3	20.0	13.3	9.3	14.0	13.3	15.2
Random, $k = 4$, L62	20.0	14.7	19.3	19.3	16.0	18.0	17.9
Regex/entity, L62	22.0	19.3	14.7	18.0	16.7	20.0	18.4